マクロ計量経済学

第6回　講義ノート
マクロ動学モデルの解法　その6

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outline

1. Why 2nd approximation to policy function is needed
2. Set up of the method
3. 1st approximation to policy function
4. 2nd approximation to policy function
   - Matlab code by Schmitt-Grohe & Uribe (2004, JEDC)
5. Example 1: Neo-classical model
   - Matlab Codes
6. Example 2: two-country RBC model
   - Matlab Codes
7. Example 3: Asset price model
   - Matlab code

Reference

1. Miao (2014) chapter 2, 2-4, and 2-5
2. Schmitt-Grohe and Uribe (2004)
   "Solving dynamic general equilibrium models using a second-order approximation to the policy function," JEDC, 28, 755-775.
3. Coeurdacier, Rey, Winant (2011)
   "The Riskey Steady State," AER, 101:3, 398-401.
4. De Groot (2013)
   "Computing the risky steady state of DSGE models," Economics Letters, 120, 566-569.

マクロ経済の動学モデルの解法

- 解法その1: 確率的ベルマン方程式　Stochastic Bellman Equation
  → 応用範囲が広いが、数値的解法で時間がかかる。
  - 状態変数の動学は、マルコフ決定過程
  - ラグランジュ方程式に置き換えることが可能。

- 解法その2: 合理的期待モデル　Rational Expectation Model
  ⇒　数値的解法での負担が少ない。
  - Uhlig (1997)
  - Blanchard & Kahn (1980)
  - Schmit-Grohe & Uribe (2004)

- 均衡解の動学式はマルコフ決定過程
  - 行列Pは　固有値が1以下で安定

$$x_t = Px_{t-1} + Qz_t, \qquad \text{均衡解}$$

$$y_t = Rx_{t-1} + Sz_t.$$

合理的期待モデル　E_t(f(y_t+1, x_t+1)) に policy function を代入

Many nonlinear rational expectations models can be represented by the following system:

Control variables　State variables

$$E_t\left[f\left(y_{t+1}, y_t, x_{t+1}, x_t\right)\right] = 0, \tag{2.27}$$

where x_t is an $(n_k + n_z) \times 1$ vector of predetermined variables and y_t is an

State variables

$n_y \times 1$ vector of non-predetermined variables in the sense of Blanchard and Kahn (1980). Let $n = n_k + n_z + n_y$ and assume the function $f : \mathbb{R}^{2n} \to \mathbb{R}^n$.

Control variables

The vector x_t consists of n_k endogenous variables and n_z exogenous shocks:

Suppose that the solution to (2.27) takes the following form:

$$y_t = g(x_t, \sigma), \quad x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1}, \tag{2.29}$$

Policy function

Dynamics of predetermined variable

where g and h are functions to be determined and η is an $n_x \times n_\varepsilon$ matrix defined by

$$\eta = \begin{bmatrix} 0 \\ \Sigma \end{bmatrix}.$$

合理的期待モデルに1st order approximationを代入

. We totally differentiate (2.27) to obtain the linearized system:

合理的期待モデルの全微分

$$f_{y'}\ E_t dy_{t+1} + f_y\ dy_t + f_{x'}\ E_t dx_{t+1} + f_x\ dx_t = 0,$$

y‘=y_t+1　　x‘=x_t+1

where the Jacobian matrix of f is evaluated at the steady state. Adopting the notation introduced in Section 1.6, we can also derive a log-linearized system. Both systems are in the form of (2.11), e.g.,

$$\begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix} \begin{bmatrix} E_t dx_{t+1} \\ E_t dy_{t+1} \end{bmatrix} = - \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} dx_t \\ dy_t \end{bmatrix}. \qquad (2.28)$$

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = 0,$$

$$y_t = g(x_t, \sigma), \qquad x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1},$$

We then define

$$\begin{aligned} 0 &= F(x, \sigma) \\ &\equiv Ef\left(g\left(h(x,\sigma) + \eta\sigma\varepsilon', \sigma\right), g(x,\sigma), h(x,\sigma) + \eta\sigma\varepsilon', x\right) \quad (2.30) \end{aligned}$$

=y_t+1 =y_t =x_t+1

We then use (2.30) to compute

$$\begin{aligned} F_\sigma(\bar{x}, 0) &= E\left\{f_{y'}\left[g_x\left(h_\sigma + \eta\varepsilon'\right) + g_\sigma\right] + f_y g_\sigma + f_{x'}\left(h_\sigma + \eta\varepsilon'\right)\right\} \\ &= f_{y'}\left[g_x h_\sigma + g_\sigma\right] + f_y g_\sigma + f_{x'} h_\sigma = 0, \end{aligned}$$

=dF/dσ

where all derivatives are evaluated at the non-stochastic steady state. This is a system of homogeneous linear equations for g_σ and h_σ. For a unique solution to exist, we must have

$$g_\sigma = h_\sigma = 0.$$

$$F_x(\bar{x}, 0) = f_{y'} g_x h_x + f_y g_x + f_{x'} h_x + f_x = 0.$$

=dF/dx

We can equivalently rewrite the above equation as

$$\begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix} h_x = - \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix}. \tag{2.31}$$

Let $A = \begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix}$, $B = - \begin{bmatrix} f_x & f_y \end{bmatrix}$, and $\hat{x}_t = x_t - \bar{x}$. Postmultiplying $\hat{x}_t$ on each side of (2.31) yields:

$$A \begin{bmatrix} I \\ g_x \end{bmatrix} h_x \hat{x}_t = B \begin{bmatrix} I \\ g_x \end{bmatrix} \hat{x}_t. \tag{2.32}$$

Using the fact that $\hat{x}_{t+1} = h_x \hat{x}_t$ and $\hat{y}_t = g_x \hat{x}_t$ for the linearized non-stochastic solution, we obtain

$$A \begin{bmatrix} \hat{x}_{t+1} \\ \hat{y}_{t+1} \end{bmatrix} = B \begin{bmatrix} \hat{x}_t \\ \hat{y}_t \end{bmatrix}. \tag{2.33}$$

1次近似(2.33)式は Blanchard-Kahnのモデルに変換できる

$$A \begin{bmatrix} \hat{x}_{t+1} \\ \hat{y}_{t+1} \end{bmatrix} = B \begin{bmatrix} \hat{x}_t \\ \hat{y}_t \end{bmatrix}. \tag{2.33}$$

Blanchard & Kahn (1980)

6.8.1 General Version

A linear model can be written (in what is known as a state space representation) as

State variables

$$B \begin{bmatrix} x_{t+1} \\ E_t y_{t+1} \end{bmatrix} = A \begin{bmatrix} x_t \\ y_t \end{bmatrix} + G\varepsilon_t, \tag{6.13}$$

Control variables

where x_t is an $(n \times 1)$ vector of predetermined variables at date t, y_t is an $(m \times 1)$ vector of non-predetermined variables at time t, $E_t y_{t+1}$ is the $(m \times 1)$ vector of expectations for the non-predetermined variables at date $t+1$, ε_t is a $(k \times 1)$ vector of stochastic shocks, A and B are $((n+m) \times (n+m))$ matrices, and G is an $((n+m) \times k)$ matrix. The difference between predetermined and non-predetermined variables is that the values of the predetermined variables at time $t+1$ do not depend on the values of the time $t+1$ shocks, while the values of the non-predetermined variables do depend on them. That is why, at

2nd order approximation

Our goal is to find an approximate solution for g and h. We shall consider local approximation around a particular point $(\bar{x}, \bar{\sigma})$. In applications, a convenient point to take is the non-stochastic steady state $(\bar{x}, \bar{\sigma})$. At this point, $f(\bar{y}, \bar{y}, \bar{x}, \bar{x}) = 0$. We then write the Taylor approximation up to the second order,

$$\begin{aligned} g(x,\sigma) &= g(\bar{x},0) + g_x(\bar{x},0)(x-\bar{x}) + g_\sigma(\bar{x},0)\sigma \\ &\quad + \frac{1}{2} g_{xx}(\bar{x},0)(x-\bar{x})^2 + g_{x\sigma}(\bar{x},0)(x-\bar{x})\sigma \\ &\quad + \frac{1}{2} g_{\sigma\sigma}(\bar{x},0)\sigma^2, \end{aligned}$$

$$\begin{aligned} h(x,\sigma) &= h(\bar{x},0) + h_x(\bar{x},0)(x-\bar{x}) + h_\sigma(\bar{x},0)\sigma \\ &\quad + \frac{1}{2} h_{xx}(\bar{x},0)(x-\bar{x})^2 + h_{x\sigma}(\bar{x},0)(x-\bar{x})\sigma \\ &\quad + \frac{1}{2} h_{\sigma\sigma}(\bar{x},0)\sigma^2, \end{aligned}$$

n_x

$n_x \times n_x$

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = 0,$$

$$y_t = g(x_t, \sigma), \qquad x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1},$$

We then define

$$\begin{aligned} 0 &= F(x, \sigma) \\ &\equiv Ef\left(g\left(h(x,\sigma) + \eta\sigma\varepsilon', \sigma\right), g(x,\sigma), h(x,\sigma) + \eta\sigma\varepsilon', x\right) \quad (2.30) \end{aligned}$$

=y_t+1 =x_t+1

We compute

$$\begin{aligned} 0 = F_{xx}(\bar{x}, 0) = &[f_{y'y} g_x h_x + f_{y'y} g_x \\ &+ f_{y'x'} h_x + f_{y'x}] g_x h_x \quad n_x \\ &+ f_{y'} g_{xx} h_x h_x + f_{y'} g_x h_{xx} \quad n_x \times n_x \\ &+ [f_{yy'} g_x h_x + f_{yy} g_x + f_{yx'} h_x + f_{yx}] g_x + f_y g_{xx} \quad n_y \times n_x \\ &+ [f_{x'y'} g_x h_x + f_{x'y} g_x + f_{x'x'} h_x + f_{x'x}] h_x + f_{x'} h_{xx} \\ &\qquad n_x \qquad n_x \times n_x \\ &+ f_{xy'} g_x h_x + f_{xy} g_x + f_{xx'} h_x + f_{xx}. \end{aligned}$$

Since we know the derivatives of f as well as the first derivatives of g and h evaluated at $(\bar{y}, \bar{y}, x, x)$, it follows that the above equation represents a system of $n \times n_x \times n_x$ linear equations for the same number of unknowns given by the elements of g_{xx} and h_{xx}.

Similarly, we compute $g_{\sigma\sigma}$ and $h_{\sigma\sigma}$ by solving the equation $F_{\sigma\sigma}(\bar{x}, 0) = 0$. We compute

$$\begin{aligned} 0 \quad = \quad & F_{\sigma\sigma}(\bar{x}, 0) = f_{y'} g_x h_{\sigma\sigma} + f_{y'y'} g_x \eta g_x \eta I \\ & + f_{y'x'} \eta g_x \eta I + f_{y'} g_{xx} \eta \eta I + f_{y'} g_{\sigma\sigma} \\ & + f_y g_{\sigma\sigma} + f_{x'} h_{\sigma\sigma} \\ & + f_{x'y'} g_x \eta \eta I + f_{x'x'} \eta \eta I. \end{aligned}$$

This is a system of n linear equations for n unknowns $g_{\sigma\sigma}$ and $h_{\sigma\sigma}$.

Finally, we show that the cross derivatives $g_{\sigma x}$ and $h_{\sigma x}$ are equal to zero when evaluated at $(\bar{x}, 0)$. We compute

$$0 = F_{\sigma x}(\bar{x}, 0) = f_{y'} g_x h_{\sigma x} + f_{y'} g_{\sigma x} h_x + f_y g_{\sigma x} + f_{x'} h_{\sigma x}.$$

This is a system of $n \times n_x$ linear homogenous equations for $n \times n_x$ unknowns $g_{\sigma x}$ and $h_{\sigma x}$. If a unique solution exists, then the solution must be equal to zero.

Matlab codes of 2nd Approximation of policy function (Smitt-Grohe and Uribe, 2004, JEDC)

Matlab code

First-order approximation
gx_hx.m

Second-order approximation
gxx_hxx.m
gss_hss.m

Obtaining the derivatives of *f*
(requires Matlab's Symbolic Math Toolbox)
anal_deriv.m
num_eval.m

Model Simulation:
simu_2nd.m

Writing the Output of anal_deriv.m to an M-file: This program saves significant amount of computational time in applications in which the output of anal_deriv.m has to be evaluated repeatedly.
anal_deriv_print2f.m by Andrea Pescatori

Example 1: The neoclassical growth model:
Contents of Matlab Codes

- neoclassical_model_run.m → main program

- neoclassical_model.m →
  description of model equations (FOC and constraints)

- neoclassical_model_ss →
  set up of parameters
  and (non-stochastic) Steady State
- neoclassical_model_simul →
  calculation of IRF, Artificial data (Monte Carlo)
  and stochastic Steady State

Main code その1

```
26
27  %  analytical 2nd order derivatives of system: f
28  [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,fyy,...
29      fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fxx,f] = neoclassical_model;
30
31  %% Numerical Evaluation
32  % Non-stochastic  Steady State and Parameter Values
33  [SIG,DELTA,ALFA,BETTA,RHO,eta,c,cp,k,kp,a,ap,A,K,C]=neoclassical_model_ss;
34
35  %Order of approximation desired
36   approx = 2;
37
38  %Obtain numerical derivatives of f
39  num_eval
40
41  %First-order approximation of Policy function
42  [gx,hx] = gx_hx(nfy,nfx,nfyp,nfxp)
```

Main code その2

```
41  %First-order approximation of Policy function
42  [gx,hx] = gx_hx(nfy,nfx,nfyp,nfxp)
43
44  %Second-order approximation of Policy function
45  [gxx,hxx] = gxx_hxx(nfx,nfxp,nfy,nfyp,nfypyp,nfypy,nfypxp,...
46                      nfypx,nfyyp,nfyy,nfyxp,nfyx,nfxpyp,...
47                      nfxpy,nfxpxp,nfxpx,nfxyp,nfxy,nfxxp,nfxx,hx,gx)
48
49  [gss,hss] = gss_hss(nfx,nfxp,nfy,nfyp,nfypyp,nfypy,nfypxp,nfypx,...
50                      nfyyp,nfyy,nfyxp,nfyx,nfxpyp,nfxpy,nfxpxp,nfxpx,...
51                      nfxyp,nfxy,nfxxp,nfxx,hx,gx,gxx,eta)
52
53
54  %% Simulation of solution
55  neoclassical_model_simul;
56
```

Example 1: The neoclassical growth model

Sec 2 of Smitt-Grohe and Uribe (2004)

$$c_t^{-\gamma} = \beta E_t c_{t+1}^{-\gamma}[\alpha A_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta],$$ Euler equation

$$c_t + k_{t+1} = A_t k_t^{\alpha} + (1-\delta)k_t,$$ Dynamics of capital

$$\ln A_{t+1} = \rho \ln A_t + \sigma \varepsilon_{t+1},$$ TFP shock process

control predetermined

for all $t \geqslant 0$, given k_0 and A_0. Let $y_t = c_t$ and $x_t = [k_t;\ \ln A_t]'$. Then

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = E_t \begin{bmatrix} y_{1t}^{-\gamma} - \beta y_{1t+1}^{-\gamma}[\alpha e^{x_{2t+1}} x_{1t+1}^{\alpha-1} + 1 - \delta] \\ y_{1t} + x_{1t+1} - e^{x_{2t}} x_{1t}^{\alpha} - (1-\delta)x_{1t} \\ x_{2t+1} - \rho x_{2t} \end{bmatrix},$$

Neoclassical_model.m

```
%(c) Stephanie Schmitt-Grohe and Martin Uribe
%Date July 17, 2001, revised 22-Oct-2004

%Define parameters
syms    SIG DELTA ALFA BETTA RHO

%Define variables
syms c cp k kp a ap

%Write equations fi, i=1:3
f1 = c + kp - (1-DELTA) * k - a * k^ALFA;
f2 = c^(-SIG) - BETTA * cp^(-SIG) * (ap * ALFA * kp^(ALFA-1) + 1 - DELTA);
f3 = log(ap) - RHO * log(a);

%Create function f
f = [f1;f2;f3];
```

define:

$$x_t = \begin{bmatrix} \ln k_t \\ \ln A_t \end{bmatrix}$$: Predetermined variables

and

$$y_t = \ln c_t.$$: Control variable

Then the non-stochastic steady-state values of y_t and x_t are, respectively:

$$\bar{y} = -0.8734.$$

and

$$\bar{x} = \begin{bmatrix} -1.7932 \\ 0 \end{bmatrix}.$$

```
29
30    % Define the vector
31    x = [k a];
32    y = c;
33    xp = [kp ap];
34    yp = cp;
35
```

```
function [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fxx,f] = neoclassical_model
```

```
35
36    %Make f a function of the logarithm of the state and contr
37    f = subs(f, [x,y,xp,yp], (exp([x,y,xp,yp])));
38    %if line 36 gives an error (whichh it will for some versic
39    %f = subs(f, [x,y,xp,yp], transpose(exp([x,y,xp,yp])));
40
41    %Compute analytical derivatives of f
42     approx = 2;  %1:1st order, 2: 2nd order
43    [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,...
44     fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,...
45     fxy,fxxp,fxx]    =  anal_deriv(f,x,y,xp,yp,approx);
46
```

このページの内容は最新ではありません。最新版の英語を参照す

subs

シンボリック代入

構文

```
snew = subs(s,old,new)
snew = subs(s,new)
snew = subs(s)
```

```
sMnew = subs(sM,oldM,newM)
sMnew = subs(sM,newM)
sMnew = subs(sM)
```

例

単一置換

この式で a を 4 に置き換えます。

```
syms a b
subs(a + b,a,4)
```

ans = $b+4$

この式で a*b を 5 に置き換えます。

```
subs(a*b^2,a*b,5)
```

ans = $5b$

```
function [fx,fxp,fy,fyp,fypyp,fypy,fypxp,...
fypx,fyyp,fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fxx]...
=anal_deriv(f,x,y,xp,yp,approx);
```

```
5   if nargin==5 %5つ目の入力が空欄のケース
6   approx=2;
7   end
8
9   nx = size(x,2);
10  ny = size(y,2);
11  nxp = size(xp,2);
12  nyp = size(yp,2);
13
14  n = size(f,1);
15
16  %Compute the first and second derivatives of f
17  fx = jacobian(f,x);
18  fxp = jacobian(f,xp);
19  fy = jacobian(f,y);
20  fyp = jacobian(f,yp);
21
```

```
22  if approx==2
23  fypyp = reshape(jacobian(fyp(:),yp),n,nyp,nyp);
24  fypy = reshape(jacobian(fyp(:),y),n,nyp,ny);
25  fypxp = reshape(jacobian(fyp(:),xp),n,nyp,nxp);
26  fypx = reshape(jacobian(fyp(:),x),n,nyp,nx);
27  fyyp = reshape(jacobian(fy(:),yp),n,ny,nyp);
28  fyy = reshape(jacobian(fy(:),y),n,ny,ny);
29  fyxp = reshape(jacobian(fy(:),xp),n,ny,nxp);
30  fyx = reshape(jacobian(fy(:),x),n,ny,nx);
31  fxpyp = reshape(jacobian(fxp(:),yp),n,nxp,nyp);
32  fxpy = reshape(jacobian(fxp(:),y),n,nxp,ny);
33  fxpxp = reshape(jacobian(fxp(:),xp),n,nxp,nxp);
34  fxpx = reshape(jacobian(fxp(:),x),n,nxp,nx);
35  fxyp = reshape(jacobian(fx(:),yp),n,nx,nyp);
36  fxy = reshape(jacobian(fx(:),y),n,nx,ny);
37  fxxp = reshape(jacobian(fx(:),xp),n,nx,nxp);
38  fxx = reshape(jacobian(fx(:),x),n,nx,nx);
39  else
40  fypyp=0; fypy=0; fypxp=0; fypx=0; fyyp=0; fyy=0;...
41      fyxp=0; fyx=0; fxpyp=0; fxpy=0; fxpxp=0;...
42      fxpx=0; fxyp=0; fxy=0; fxxp=0; fxx=0;
43  end
```

reshape

R2024b

既存の要素を再配列して配列を形状変更する

ページ内をすべて折りたたむ

構文

```
B = reshape(A,sz)
B = reshape(A,sz1,...,szN)
```

例

ベクトルを行列に形状変更

1 行 10 列のベクトルを 5 行 2 列の行列に形状変更します。

```
A = 1:10;
B = reshape(A,[5,2])
```

```
B = 5×2

     1     6
     2     7
     3     8
     4     9
     5    10
```

1次近似　gx_hx()

```
function [gx,hx,exitflag]=gx_hx(fy,fx,fyp,fxp,stake);

if nargin<5
    stake=1;
end
exitflag = 1;

%Create system matrices A,B
A = [-fxp -fyp];
B = [fx fy];
NK = size(fx,2);

%Complex Schur Decomposition
[s,t,q,z] = qz(A,B);

%Pick non-explosive (stable) eigenvalues
slt = (abs(diag(t))<stake*abs(diag(s)));
nk=sum(slt);

%Reorder the system with stable eigs in upper-left
[s,t,q,z] = ordqz(s,t,q,z,slt);

%Split up the results appropriately
z21 = z(nk+1:end,1:nk);
z11 = z(1:nk,1:nk);

s11 = s(1:nk,1:nk);
t11 = t(1:nk,1:nk);
```

```
%Identify cases with no/multiple solutions
if nk>NK
    warning('The Equilibrium is Locally Indeterminate')
    exitflag=2;
elseif nk<NK
    warning('No Local Equilibrium Exists')
    exitflag = 0;
end

if rank(z11)<nk;
    warning('Invertibility condition violated')
    exitflag = 3;
end

%Compute the Solution
z11i = z11\eye(nk);
gx = real(z21*z11i);
hx = real(z11*(s11\t11)*z11i);
```

2次近似

```
23
24  function [gxx,hxx] = gxx_hxx(fx,fxp,fy,fyp, ...
25      fypyp,fypy,fypxp,fypx,fyyp,fyy,fyxp, ...
26      fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fxx,hx,gx)
27
28  m=0;
29  nx = size(hx,1); %rows of hx and hxx
30  ny = size(gx,1); %rows of gx and gxx
31  n = nx + ny; %length of f
32  ngxx = nx^2*ny; %elements of gxx
33
34  sg = [ny nx nx]; %size of gxx
35  sh = [nx nx nx]; %size of hxx
36
37  Q = zeros(n*nx*(nx+1)/2,n*nx*nx);
38  q = zeros(n*nx*(nx+1)/2,1);
39  gxx=zeros(sg);
40  hxx=zeros(sh);
41  GXX=zeros(sg);
42  HXX=zeros(sh);
43
```

```
45  for i=1:n
46  for j=1:nx
47  %for k=1:nx
48  for k=1:j
49  m = m+1;
50
51  %First Term
52  q(m,1) = ( shiftdim(fypyp(i,:,:),1) * gx * hx(:,k) ...
53      + shiftdim(fypy(i,:,:),1) * gx(:,k) + shiftdim(fypxp(i,:,:),1) ...
54      *  hx(:,k) + shiftdim(fypx(i,:,k),1) )' * gx * hx(:,j);
55
56  % Second term
57  GXX(:) = kron(ones(nx^2,1),fyp(i,:)');
58  pGXX = permute(GXX,[2 3 1]);
59  pGXX(:) = pGXX(:) .* kron(ones(nx*ny,1),hx(:,j));
60  GXX=ipermute(pGXX,[2 3 1]);
61  pGXX = permute(GXX,[3 1 2]);
62  pGXX(:) = pGXX(:) .* kron(ones(nx*ny,1),hx(:,k));
63  GXX=ipermute(pGXX,[3 1 2]);
64  Q(m,1:ngxx)=GXX(:)';
65  GXX=0*GXX;
```

```
67  %Third term
68  HXX(:,j,k) = (fyp(i,:) * gx)';
69  Q(m,ngxx+1:end)=HXX(:)';
70  HXX = 0*HXX;
71
72  %Fourth Term
73  q(m,1) = q(m,1) + ( shiftdim(fyyp(i,:,:),1) * gx * hx(:,k) ...
74      +  shiftdim(fyy(i,:,:),1) * gx(:,k) + shiftdim(fyxp(i,:,:),1) ...
75      * hx(:,k) +  shiftdim(fyx(i,:,k),1) )' * gx(:,j);
76
77  %Fifth Term
78  GXX(:,j,k)=fy(i,:)';
79  Q(m,1:ngxx) = Q(m,1:ngxx) + GXX(:)';
80  GXX = 0*GXX;
81
82  %Sixth term
83  q(m,1) = q(m,1) + ( shiftdim(fxpyp(i,:,:),1) ...
84      * gx * hx(:,k) + shiftdim(fxpy(i,:,:),1) ...
85      * gx(:,k) + shiftdim(fxpxp(i,:,:),1) ...
86      * hx(:,k) + fxpx(i,:,k)')' * hx(:,j);
87
88  %Seventh Term
89  HXX(:,j,k)=fxp(i,:)';
90  Q(m,ngxx+1:end) = Q(m,ngxx+1:end) + HXX(:)';
91  HXX = 0*HXX;
92
```

```
92
93  %Eighth Term
94  q(m,1) = q(m,1) +  shiftdim(fxyp(i,j,:),1)
95  * gx * hx(:,k) +  shiftdim(fxy(i,j,:),1)
96  * gx(:,k) +  shiftdim(fxxp(i,j,:),1) * hx(:,k) + fxx(i,j,k);
97
98  end %k
99  end %j
100 end %i
101
102 A = temp(nx,ny);
103
104 Qt = Q* A;
105
106 xt = -Qt\q;
107 x = A* xt;
108
109 gxx(:)=x(1:ngxx);
110 hxx(:) = x(ngxx+1:end);
111
```

The coefficients of the linear terms are:

$$g_x = [0.2525 \quad 0.8417]$$

and

$$h_x = \begin{bmatrix} 0.4191 & 1.3970 \\ 0.0000 & 0.0000 \end{bmatrix}.$$

The coefficients of the quadratic terms are given by:

$$g_{xx}(:,:,1) = [-0.0051 \quad -0.0171],$$

$$g_{xx}(:,:,2) = [-0.0171 \quad -0.0569]$$

and

$$h_{xx}(:,:,1) = \begin{bmatrix} -0.0070 & -0.0233 \\ 0 & 0 \end{bmatrix},$$

$$h_{xx}(:,:,2) = \begin{bmatrix} -0.0233 & -0.0778 \\ 0 & 0 \end{bmatrix}.$$

Finally, the coefficients of the quadratic terms in σ are:

$$g_{\sigma\sigma} = -0.1921$$

and

$$h_{\sigma\sigma} = \begin{bmatrix} 0.4820 \\ 0 \end{bmatrix}.$$

$$\begin{aligned} g(x,\sigma) = {} & g(\bar{x},0) + g_x(\bar{x},0)(x-\bar{x}) + g_\sigma(\bar{x},0)\sigma \\ & + \frac{1}{2}g_{xx}(\bar{x},0)(x-\bar{x})^2 + g_{x\sigma}(\bar{x},0)(x-\bar{x})\sigma \\ & + \frac{1}{2}g_{\sigma\sigma}(\bar{x},0)\sigma^2, \end{aligned}$$

Matlabの計算結果

```
gx =

    0.2525    0.8417


hx =

    0.4191    1.3970
         0         0


gxx(:,:,1) =

   -0.0051   -0.0171


gxx(:,:,2) =

   -0.0171   -0.0569
```

```
hxx(:,:,1) =

   -0.0070   -0.0233
         0         0


hxx(:,:,2) =

   -0.0233   -0.0778
         0         0


gss =

   -0.1921


hss =

    0.4820
         0
```

Policy function

$$y_t = g(x_t, \sigma),$$

Then, the laws of motion of these two variables are given by

$$\hat{c}_t = 0.2525\hat{k}_t + 0.8417\hat{A}_t + \frac{1}{2}[-0.0051\hat{k}_t^2 - 0.0341\hat{k}_t\hat{A}_t - 0.0569\hat{A}_t^2 - 0.1921\sigma^2]$$

and

$$\hat{k}_{t+1} = 0.4191\hat{k}_t + 1.3970\hat{A}_t$$

$$x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1},$$

$$+\frac{1}{2}\left[-0.0070\hat{k}_t^2 - 0.0467\hat{k}_t\hat{A}_t - 0.0778\hat{A}_t^2 + 0.4820\sigma^2\right].$$

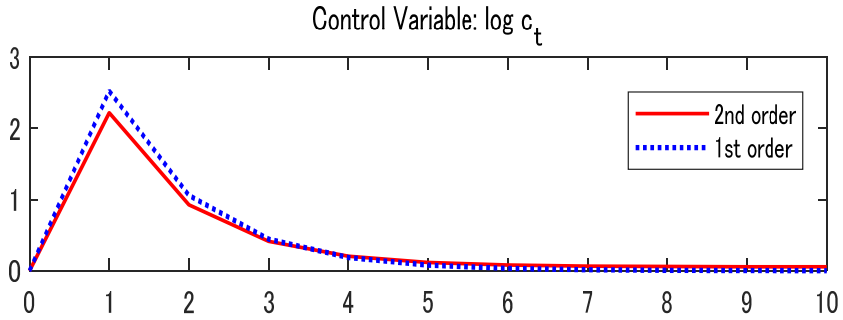
It can be verified that these numbers coincide with those obtained by Sims (2000).

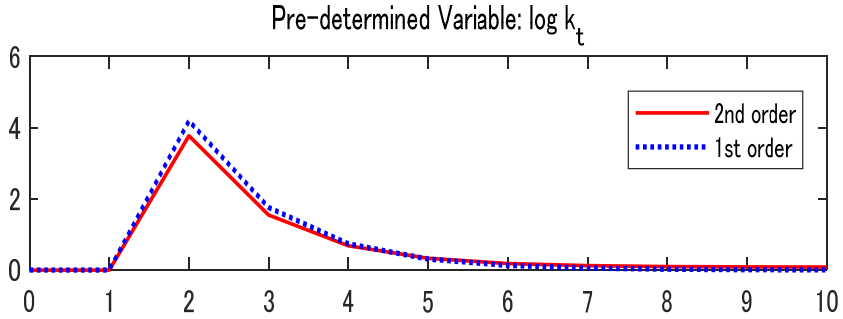
Neoclassical_simul.m

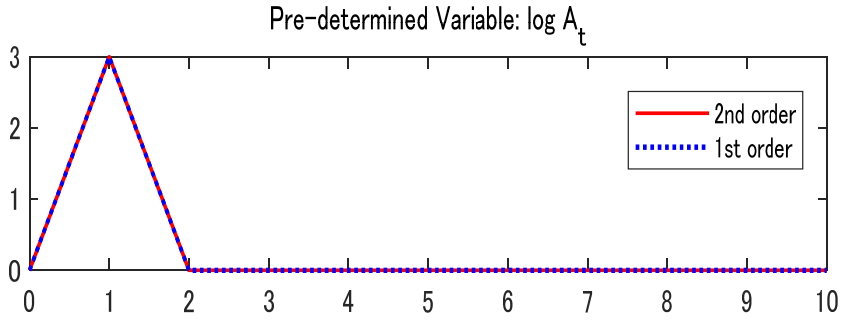
```
%% Check Solution
% cal   unconditional_mean
sig = 1;  % St Dev of shock
[Ey,Ex] = unconditional_mean(gx, hx, gxx, hxx, gss, hss, eta, sig)

x0 = Ex;
e = [ 0 0; zeros(200,2)];
[Y02,X02] = simu_2nd(gx, hx, gxx, hxx, gss, hss, eta, sig, x0, e);
x_ss1 = X02(end,:)' % stochastic steady state

% cal stochastic steady state by fixed point
x_d = zeros(4,1);  % x_d = zeros(4*2,1);
 x0_ss = csolve('eq_stochastic_steady_state',x_d ,[],1e-5,2000,hx, hxx, hss, sig )
%   x0_ss2   = x0_ss(1:4,1)+x0_ss(5:8,1)
[Error] = eq_stochastic_steady_state(x0_ss , hx, hxx, hss, sig )

```

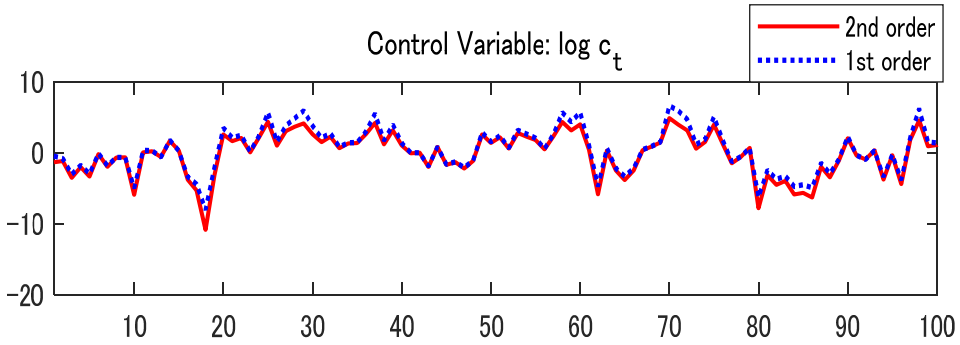
```
% Impulse Response Function
% sig = 0.1;           % St Dev of shock
x0_1 = [0;0;0;0];        % initial values of predetermined variables
x0_2 = x_ss1;
h = 40;         % horizon of IRF
e = [ 1 0; zeros(h,2)]; % generations of shocks
t = 0:1:h+1;

[Y2,X2] = simu_2nd(gx, hx, gxx, hxx, gss, hss, eta, sig, x0_2, e);
[Y1,X1] = simu_1st(gx, hx, eta, sig, x0_1, e);

figure('Name','IRF of Kim Model')
subplot(3,1,1)
     plot(t,Y2-Y2(1,1),'r-','LineWidth',1.5);
   hold on
     plot(t,Y1-Y1(1,1),'b:','LineWidth',1.5);
   hold off
     title('Control Variable: log c_t')
     legend('2nd order', '1st order')
     xlim([0 h])
%      ylim( [-0.025 0.025])
```

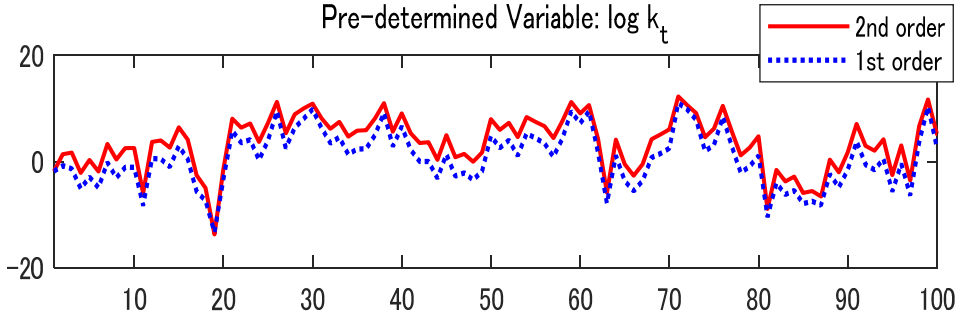
インパルス応答の1次近似と2次近似の比較

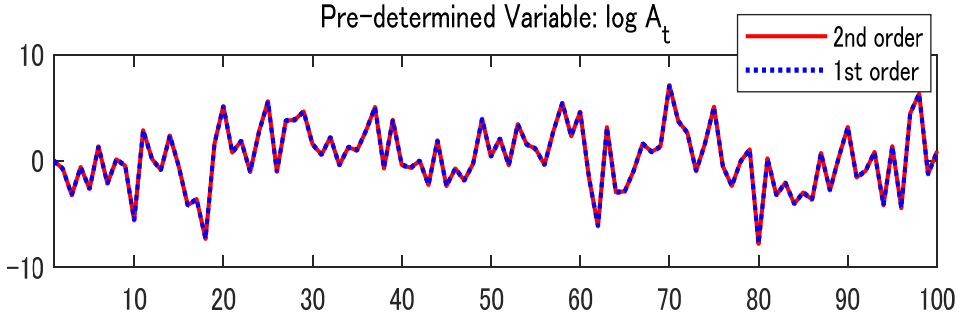












Example 2: A two-country neoclassical model:
Contents of Matlab Codes

- kim_model_run.m → main program
- kim_model.m →
  description of model equations (FOC and constraints)
- kim_model_ss →
  set up of parameters
  and (non-stochastic) Steady State
- kim_model_simul →
  calculation of IRF, Artificial data (Monte Carlo)
  and stochastic Steady State

Example 2: A two-country neoclassical model with complete asset markets

The planner's objective function is given by

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_{1t}^{1-\gamma} - 1}{1-\gamma} + \frac{C_{2t}^{1-\gamma} - 1}{1-\gamma} \right],$$

where C_{it}, $i = 1, 2$, denotes consumption of the representative household of country i in period t. The planner maximizes this utility function subject to the budget constraint

$$C_{1t} + C_{2t} + K_{1t+1} - (1-\delta)K_{1t} + K_{2t+1} - (1-\delta)K_{2t} = A_{1t}K_{1t}^{\alpha} + A_{2t}K_{2t}^{\alpha},$$

where K_{it} denotes the stock of physical capital in country i and A_{it} is an exogenous technology shock whose law of motion is given by

$$\ln A_{it} = \rho_i \ln A_{it-1} + \sigma \varepsilon_{it}, \qquad i = 1, 2,$$

where $\varepsilon_{it} \sim NIID(0, 1)$ and $\rho_i \in (-1, 1)$. The optimality conditions associated with this problem are the above period-by-period budget constraint and

$$C_{1t} = C_{2t},$$

$$C_{1t}^{-\gamma} = \beta E_t C_{1t+1}^{-\gamma} [\alpha A_{1t+1} K_{1t+1}^{\alpha-1} + (1-\delta)],$$

$$C_{1t}^{-\gamma} = \beta E_t C_{1t+1}^{-\gamma} [\alpha A_{2t+1} K_{2t+1}^{\alpha-1} + (1-\delta)].$$

kim_model.m

```
%Define parameters
syms GAMA DELTA ALFA RHO BETTA

%Define variables
syms c cp k1 k1p k2 k2p a1 a1p a2 a2p
%Write equations (e1 e2 e3 e4 e5)
e1 = c^(-GAMA) - BETTA * cp^(-GAMA) * (ALFA * a1p * k1p^(ALFA-1) + 1 - DELTA);

e2 = c^(-GAMA) - BETTA * cp^(-GAMA) * (ALFA * a2p * k2p^(ALFA-1) + 1 - DELTA);

e3 = 2 * c + k1p - (1-DELTA) * k1 + k2p - (1-DELTA) * k2 - a1 * k1^(ALFA) - a2 * k2^(ALFA);

e4 = log(a1p) - RHO * log(a1);

e5 = log(a2p) - RHO * log(a2);

%Create function f
f = [e1;e2;e3;e4;e5];
```

We use the following parameter values: $\gamma = 2$; $\delta = 0.1$; $\alpha = 0.3$; $\rho = 0$; and $\beta = 0.95$.

Given this parameterization, the second-order approximation to the policy function is given by:

$$\hat{K}_{1t+1} = [\,0.4440 \quad 0.4440 \quad 0.2146 \quad 0.2146\,] \begin{bmatrix} \hat{K}_{1t} \\ \hat{K}_{2t} \\ \hat{A}_{1t} \\ \hat{A}_{2t} \end{bmatrix}$$

$$+ \frac{1}{2}[\,\hat{K}_{1t} \quad \hat{K}_{2t} \quad \hat{A}_{1t} \quad \hat{A}_{2t}\,]$$

$$\times \begin{bmatrix} 0.22 & -0.18 & -0.023 & -0.088 \\ -0.18 & 0.22 & -0.088 & -0.023 \\ -0.023 & -0.088 & 0.17 & -0.042 \\ -0.088 & -0.023 & -0.042 & 0.17 \end{bmatrix} \begin{bmatrix} \hat{K}_{1t} \\ \hat{K}_{2t} \\ \hat{A}_{1t} \\ \hat{A}_{2t} \end{bmatrix}$$

$$- \frac{1}{2} 0.166\sigma^2,$$

$$\hat{K}_{2t+1} = \hat{K}_{1t+1},$$

$$[h(x,\sigma)]^j = [h(\bar{x},0)]^j + [h_x(\bar{x},0)]^j_a [(x-\bar{x})]_a + [h_\sigma(\bar{x},0)]^j [\sigma]$$

$$+ \frac{1}{2}[h_{xx}(\bar{x},0)]^j_{ab}[(x-\bar{x})]_a[(x-\bar{x})]_b$$

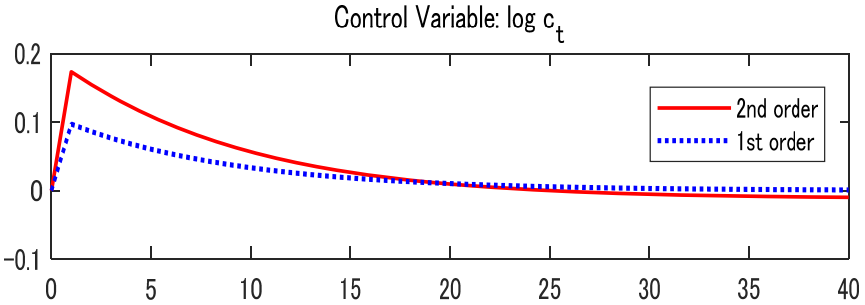
$$+ \frac{1}{2}[h_{x\sigma}(\bar{x},0)]^j_a[(x-\bar{x})]_a[\sigma]$$

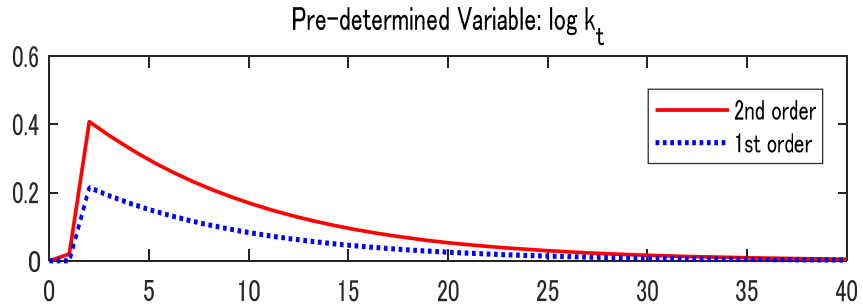
$$+ \frac{1}{2}[h_{\sigma x}(\bar{x},0)]^j_a[(x-\bar{x})]_a[\sigma]$$

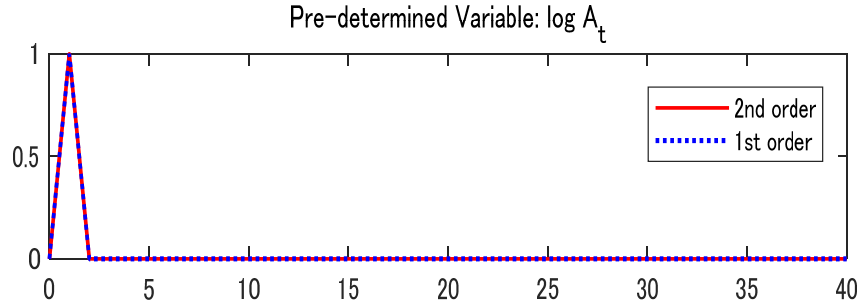
$$+ \frac{1}{2}[h_{\sigma\sigma}(\bar{x},0)]^j[\sigma][\sigma],$$

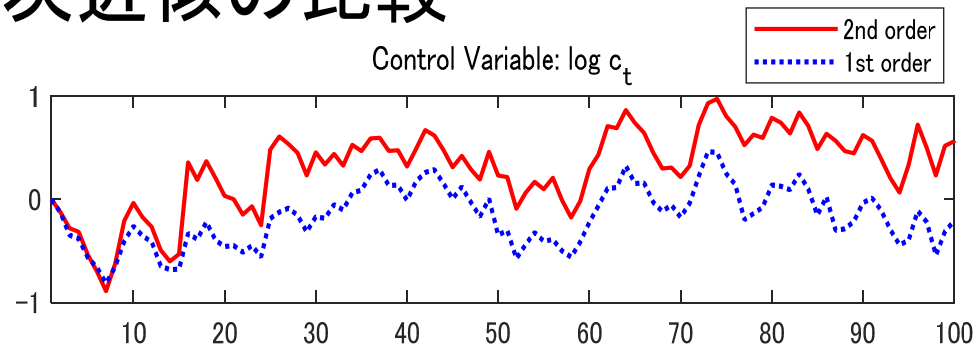
```
gx =

    0.2013    0.2013    0.0973    0.0973


hx =

    0.4440    0.4440    0.2146    0.2146
    0.4440    0.4440    0.2146    0.2146
         0         0         0         0
         0         0         0         0


gxx(:,:,1) =

    0.1013   -0.0796   -0.0093   -0.0385


gxx(:,:,2) =

   -0.0796    0.1013   -0.0385   -0.0093


gxx(:,:,3) =

   -0.0093   -0.0385    0.0787   -0.0186


gxx(:,:,4) =

   -0.0385   -0.0093   -0.0186    0.0787


hxx(:,:,1) =

    0.2178   -0.1812   -0.0232   -0.0876
    0.2178   -0.1812   -0.0232   -0.0876
         0         0         0         0
         0         0         0         0


hxx(:,:,2) =

   -0.1812    0.2178   -0.0876   -0.0232
   -0.1812    0.2178   -0.0876   -0.0232
         0         0         0         0
         0         0         0         0
```

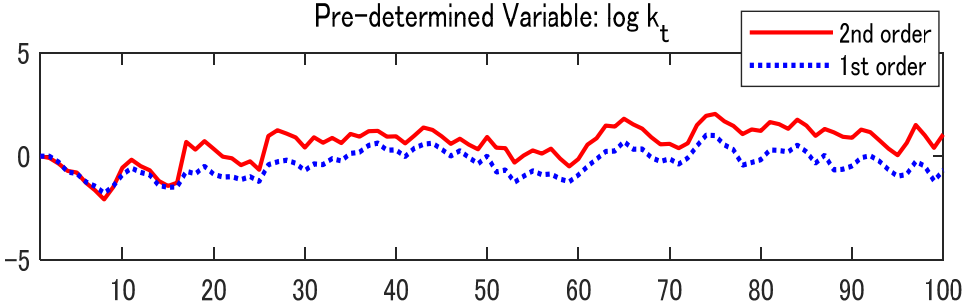
インパルス応答の1次近似と2次近似の比較

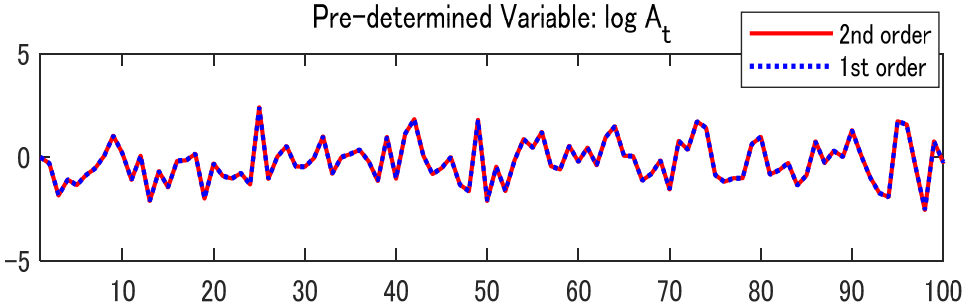












Example 3: An asset pricing model:
Contents of Matlab Codes

- asset_run.m → main program

- asset_model.m →
  description of model equations (FOC and constraints)

- asset_ss →
  set up of parameters
  and (non-stochastic) Steady State

- asset_model_simul →
  calculation of IRF, Artificial data (Monte Carlo)
  and stochastic Steady State

Example 3: An asset pricing model

Consider the following endowment economy analyzed by Burnside (1998) and Collard and Juillard (2001b). The representative agent maximizes the lifetime utility function

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{\theta}}{\theta},$$

subject to

$$p_t e_{t+1} + C_t = p_t e_t + d_t e_t$$

and a borrowing limit that prevents agents from engaging in Ponzi games. In the above expressions, C_t denotes consumption, p_t the relative price of trees in terms of consumption goods, e_t the number of trees owned by the representative household at the beginning of period t, and d_t the dividends per tree in period t. Dividends are

assumed to follow an exogenous stochastic process given by

$$d_{t+1} = e^{x_{t+1}} d_t,$$

where e^{x_t} denotes the gross growth rate of dividends. The natural logarithm of the gross growth rate of dividends, x_t, is assumed to follow an exogenous AR(1) process

$$x_{t+1} = (1 - \rho)\bar{x} + \rho x_t + \sigma\eta\varepsilon_{t+1},$$

and $\varepsilon_t \sim NIID(0, 1)$.[14]

The optimality conditions associated with the household's problem are the above budget constraint, the borrowing limit, and

$$p_t C_t^{\theta-1} = \beta E_t C_{t+1}^{\theta-1}(p_{t+1} + d_{t+1}).$$

Burnside (1998) shows that the non-explosive solution to this equation is of the form

$$y_t \equiv g(x_t, \sigma) = \sum_{i=1}^{\infty} \beta^i \mathrm{e}^{a_i + b_i(x_t - \bar{x})}, \tag{9}$$

where

$$a_i = \theta \bar{x} \mathrm{i} + \frac{\theta^2 \sigma^2 \eta^2}{2(1-\rho)^2} \left[\mathrm{i} - \frac{2\rho(1-\rho^i)}{1-\rho} + \frac{\rho^2(1-\rho^{2i})}{1-\rho^2} \right]$$

and

$$b_i = \frac{\theta \rho (1-\rho^i)}{1-\rho}.$$

Asset_model.m

```
13    %Define parameters
14    syms XBAR RHO THETA BETTA
15
16    %Define variables
17    syms p pp a ap
18
19    %Write equations (e1 and e2)
20    e1 = -p + BETTA * (1 + pp) * exp(THETA * ap);
21
22    e2 = -ap + (1-RHO) * XBAR + RHO * a;
23
24    %Create function f
25    f = [e1;e2];
26
```

A second-order approximation of (9) around $x_t = \bar{x}$ and $\sigma = 0$ yields

$$y_t \approx g(\bar{x}, 0) + g_x(x_t - \bar{x}) + \frac{1}{2} g_{xx}(x_t - \bar{x})^2 + \frac{1}{2} g_{\sigma\sigma}\sigma^2, \tag{10}$$

where

$$g_x = \frac{\theta\rho\beta e^{\theta\bar{x}}}{(1 - \beta e^{\theta\bar{x}})(1 - \beta e^{\theta\bar{x}}\rho)},$$

$$g_{xx} = \left(\frac{\rho\theta}{1-\rho}\right)^2 \left[\frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}} - \frac{2\beta e^{\theta\bar{x}}\rho}{1 - \beta e^{\theta\bar{x}}\rho} + \frac{\beta e^{\theta\bar{x}}\rho^2}{1 - \beta e^{\theta\bar{x}}\rho^2}\right],$$

and

$$g_{\sigma\sigma} = \left(\frac{\theta\eta}{1-\rho}\right)^2 \left[\frac{\beta e^{\theta\bar{x}}}{(1 - \beta e^{\theta\bar{x}})^2} + \left(\frac{\rho^2}{1-\rho^2} - \frac{2\rho}{1-\rho}\right) \times \frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}} + \frac{2\rho^2}{1-\rho}\frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}\rho} - \frac{\rho^4}{1-\rho^2}\frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}\rho^2}\right].$$

We follow Burnside (1998) and Collard and Juillard (2001b) and use the calibration $\beta = 0.95$, $\theta = -1.5$, $\rho = -0.139$, $\bar{x} = 0.0179$, and $\eta = 0.0348$. Then, evaluating the above expressions we obtain

$$y_t \approx 12.30 + 2.27(x_t - \bar{x}) + \frac{1}{2} 0.42(x_t - \bar{x})^2 + \frac{1}{2} 0.35\sigma^2.$$

```
>> main_asset_model
gx =
    2.2731

hx =
   -0.1390

gxx =
    0.4205

hxx =
     0

gss =
    0.3507

hss =
     0

Ey =
    0.1756

Ex =
     0

x0_ss =
     0
```

インパルス応答の1次近似と2次近似の比較